Sum to n

The sumTo function adds integers from 1 through a nonnegative n. Its base case is sumTo(0) = 0. Otherwise return n + sumTo(n - 1). For n = 5, the result is 15.

Counting decimal digits

For a nonnegative integer, countDigits returns 1 if n is less than 10. This includes zero, which has one decimal digit. Otherwise return 1 + countDigits(n / 10), using integer division to remove the last digit.

Ascending recursive printing

To print 1 through n in ascending order, stop when n is 0. First call printOneToN(n - 1), then print n. Printing before the recursive call would reverse the order for positive n.

Recursive integer power

For a nonnegative exponent, power(x, 0) returns 1. Otherwise return x * power(x, n - 1). This version makes one recursive call per exponent step and takes linear time in n. Inputs must keep the integer result in range.

Counting Hanoi moves

The minimum Hanoi move count for zero disks is 0. For n positive disks, moves(n) = 2 * moves(n - 1) + 1, giving 2^n - 1 moves. Three disks require 7 moves. Computing this count makes one recursive call per level; actually printing all moves is a different operation.